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Adaptive immunity affects learning behavior in mice

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Abstract

Regulation of neuronal plasticity by the immune system is an evolving field of modern neuroscience. Here we employ immune deficient mice to examine the role of the immune system in learning behavior of mice in a variety of cognitive tasks. While no motivation or motor function deficits are evident in severe combined immune deficient (scid) mice, there was significant impairment in acquisition of cognitive tasks as compared to wild-type (WT) control mice. Moreover, acute depletion of adaptive immunity in adult WT mice significantly impaired learning behavior. Passive transfer of autologous T cells into WT mice following ablation of adaptive immunity restored previously impaired cognitive function. These results suggest that throughout lifetime, immune system supports cognitive function and may therefore have far-reaching therapeutic implications for cognitive disorders.
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Keywords: T cells; Cognition; Radial arm water maze; Morris water maze; Barnes maze; Open field; Bone marrow transplantation

1. Introduction

The immune response enables defense against pathogens, environmental toxins, and trauma. While immune response in neural tissue is restricted due to the blood–brain barrier (Carson et al., 2006; Wekerle, 2006), these restrictions can be overcome and do not always exist (Aloisi et al., 2001; Pachter et al., 2003; Reinke and Fabry, 2006). Thus, for example, passive transfer of T cells directed against CNS autoantigens reduces the post-traumatic loss of neurons and functional defects (Hauben et al., 2000; Moalem et al., 1999).

Recent studies have shown that cognitive abilities of mice with severe combined immune deficiency (scid) are inferior to those of the matched wild-type (WT) controls (Kipnis et al., 2004). Although the difference in learning behavior between WT and scid mice was robust and reproducible, previous studies did not address the nature of cognitive difference between the two strains. Moreover, these studies did not rule out a possibility of reduced motivation or impairment in sensorimotor functions of scid mice.

The current study was designed to better understand which parts of a complex process of learning are affected or controlled by the immune system. Moreover, we consider a causal relationship between absence of T cells and loss of cognitive function by examining the cognitive performance of adult mice after subjecting them to ablation of their adaptive immunity.

2. Materials and methods

2.1. Animals

Inbred male adult (8–12 weeks old) Balb/c/BySmn and CBySmn.CB17-*Prkdc scid*/J mice were purchased from Jackson Laboratories (Bar Harbor, ME). The animals were housed in light- and temperature-controlled rooms and matched for age in each experiment. Both strains were kept in identical housing conditions. The life span of scid mice is comparable to that of WT mice and there are no specific dietary requirements for scid mice. Animals that were examined in the open field, were given at least a week of rest before were tested for one of the cognitive

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tasks. None of the groups were tested for more than one cognitive task. At the last day of each task performance animals were euthanized. All procedures complied with regulations of the Institutional Animal Care and Use Committee (IACUC) at University of Nebraska Medical Center.

2.2. Morris water maze test

Mice were given four trials per day, for 4 consecutive days, to find a hidden platform located one cm below the water surface in a pool 1 m in diameter. The water temperature was kept in all experiments at 22–23 °C. Within the testing room only distal visuo-spatial cues were available to the mice for location of the submerged platform. The Balb/c mice used in this study have impaired vision and can distinguished light from darkness but cannot fully see the objects. To address this we used lights as visual cues and for the visible trial we attached a glowing stick to the platform. Changing the objects on the walls to lights and adding the glowing stick significantly improved the learning ability of Balb/c mice in this task. The escape latency, i.e., the time required by the mouse to find and climb onto the platform, was recorded for up to 60 s. Each mouse was allowed to remain on the platform for 20 s, and was then moved from the maze to its home cage. If the mouse did not find the platform within 60 s, it was manually placed on the platform and returned to its home cage after 20 s. The inter-trial interval for each mouse was 30 s. On day 5 the platform was removed from the pool, and each mouse was tested by a probe trial for 60 s. On days 6 and 7 the platform was placed at the opposite quadrant, and the mouse was retrained in four sessions each day. On day 8 the platform was placed in a quadrant different from acquisition or reversal training quadrants and was raised 2 cm above the water and had a bright colored stick on it to make it better visible (visible platform). Data were recorded using an EthoVision automated tracking system (Noldus Information Technology, The Netherlands). Statistical analysis was performed using ANOVA and individual points were compared using a post-hoc Bonferroni test.

2.3. Open field test

Motor activity level was measured in open field test. A square arena (50 × 50 cm) was placed in the field of a recording camera and lit by the ceiling light. Behavior was registered and analyzed with Ethovision video tracking system. After the mouse was placed in the center of the arena it was given 5 min to explore the environment.

2.4. Barnes maze test

Hippocampal dependent spatial learning ability was tested in Barnes Maze test (Barnes and Eppich, 1989). The Barnes maze apparatus was purchased from San Diego Instruments, San Diego, CA. An escape box was placed underneath one of the holes and its location was constant for each mouse. Environmental cues (posters on the walls, shelves, refrigerator, computer and experimenter) were placed around the room to use during task performance. Bright ceiling light directed to the maze surface was used as an aversive stimulus to increase escape motivation of animals. Ethovision video tracking system was used for behavior registration and analysis.

At the beginning of each trial, the mouse was placed in the starting tube (8 inch high, 6 inch in diameter) and allowed 15 s for adaptation to new conditions. After starting tube removal, the mouse was allowed to explore the maze for 2 min after which it was placed in the box manually. In order to reduce olfactory cues, the maze surface was sprayed and wiped with 70% ethanol between trials. Each mouse had a block of four trials per day with 10-min interval between trials for 5 days. Escape latency, pathway length and speed were measured. Statistical analysis was performed using ANOVA and individual points were compared using a post-hoc Bonferroni test.

2.5. Radial arm water maze test

Mice were tested for two days on 6-radial-arm water-maze paradigm (Gordon et al., 2001; Wilcock et al., 2006) in a water pool with 1 m diameter. The water temperature was kept in all experiments at 21 °C. On day one, 15 trials were run in five blocks of 3 trials each. Mice were labeled and the observing investigators were blind to the identity of mice. All of the examined mice in each experiment were randomly divided into small sub-groups of 3–4 mice in each. Each subgroup had a long rest period before running the following 3-trial block. The start arm varied in each trial, and the goal arm remained constant for a given individual for both days. To minimize odor cues the goal arm was changed for each mouse in a subgroup. On day one the platform was hidden for the first 12 trials and visible during the last three trials. On day 2 the platform was hidden for all trials. Errors of incorrect arm entries and latency (time to find the platform; if not found by end of the trial latency of 60 s was assigned) were measured in 1-min time periods (or increments). If a mouse failed to make a choice for 20 s an error of one was assigned. Results were analyzed by two-way ANOVA and individual points were compared using a post-hoc Bonferroni test.

2.6. Purification of murine $CD3^+$ T cells

Lymph nodes (axillary, inguinal, superficial cervical, mandibular, and mesenteric) were harvested and mashed. T cells were purified (enriched by negative selection) on T-cell columns (R&D Systems, Minneapolis, MN). A sample was labeled with CD3/TcR β chain fluorescent antibodies and analyzed by FACS for purity. The population contained 98% $CD3^+$ T cells (more than 99% positive for TcR). Each mouse was injected 20×10^6 cells i.v.

2.7. BM isolation

Mice were sacrificed using CO_2 and saturated with 70% alcohol. Skin was removed from the lower part of the body. Tissue was removed from legs with scissors and dissected away from body. Remaining tissue was cleaned from the pelvic and femoral bones and separated at knee joint. Bone ends were cut off and filled with PBS containing 10% fetal calf serum using 10 cc syringe with 25 g needle. Cells were centrifuged and re-suspended at the concentration of 10×10^6 cells/ml (200 μl/mouse were injected i.v.).

2.8. Irradiation and BM replenishment

Adult WT Balb/c/BySmn mice were subjected to a lethal split-dose gamma-irradiation (300 rad followed 2 days later by 950 rad). Three hours after the second irradiation mice were injected with 2×10^6 bone-marrow cells freshly isolated from identical WT mice or from scid mice (CBySmn.CB17-*Prkdcscid*/J).

2.9. Statistical analysis

For statistical analysis Two-way repeated measures ANOVA was used to compare results obtained with Morris Water maze, Radial arm water maze, and Barnes maze. Bonferroni post-hoc tests were used for individual pair comparison.

3. Results

In this study we assessed spatial learning and memory of WT and scid mice using numerous cognitive tasks. First, we performed the classical Morris water maze (MWM) as was previously published. WT (Balb/c/BySmn) and scid mice (CBySmn.CB17-Prkdc-scid/J) on an identical genetic background were used in all experiments. During the first 4 days of MWM, scid mice showed a significant impairment in task acquisition. At the end of the fourth day scid mice were able to find the platform within 30 s as compared to

10 s required for WT mice to accomplish an identical task (Fig. 1A).

We examined retrieval of an acquired memory using a probe trial of MWM (Sajadi et al., 2006). When both groups were tested in a probe trial on the 5th day no difference between the groups was observed (Fig. 1B), suggesting that the memory is not dependent on adaptive immunity. In the reversal trial, when the platform location was changed to an opposite quadrant, immune deficient mice exhibited again significantly impaired performance (Fig. 1C). To eliminate the possibility of a lack of motivation in scid mice, a visible test was performed on the 8th day. No difference was observed between the two groups of mice in performance of visible trials (Fig. 1D), suggesting that scid mice did not differ in motivation to escape from the water. Velocity for the duration of experiment was measured and no difference between the groups was found (Fig. 1E).

To examine activity of scid vs. WT mice we tested their behavior in the open field using naïve (previously not exposed to any cognitive task) mice. No major differences between the two groups were found (Fig. 1F). The subtle differences were in the average time spent in the center by scid mice and their latency of the first center crossing. Overall, however, center crossings and the distance moved as well as the mean velocity were comparable for both groups.

Another group of scid and WT mice was examined on the rotarod and no difference between the two groups was found as well (Fig. 1G), further substantiating normal motor function in scid mice.

To substantiate the data obtained in MWM and to rule out that the observed effect is due to differential stress-related behavior of the two strains, we examined learning behavior of scid mice in a water-free Barnes maze. Time latency for WT mice to reach the hidden box was significantly lower than for scid mice (Fig. 2A). The total distance to reach the escape box was also significantly shorter for WT mice than for scid mice from day 3 and onward (Fig. 2B). Representative micrographs of paths are presented (Fig. 2C). No difference in velocity between the groups was observed (Fig. 2D).

Another spatial learning and memory task that is widely used in Alzheimer's research is a six-radial arm water maze (RAWM) (Wilcock et al., 2006). This is an extensive 2-day long training task consisting of a visible and invisible platform trials. Comparison of the RAWM task performance of WT and scid mice disclosed a significantly longer escape latency for the scid mice (Fig. 3A) and significantly more errors in their attempts to reach the hidden platform (Fig. 3B).

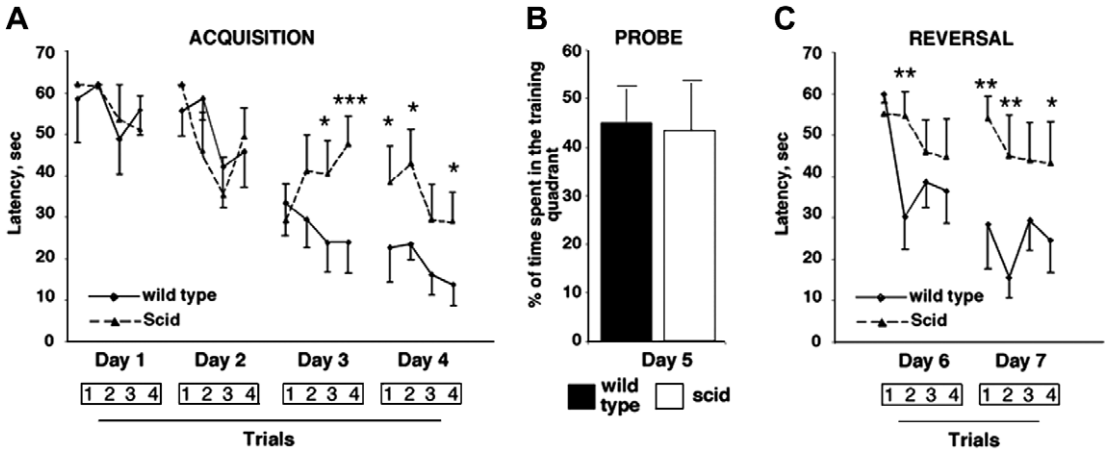
Previous studies with MWM showed that impaired learning of T cell-deficient mice can be reversed by injection of T cells from genetically-identical WT donors (Kipnis et al., 2004). To verify that T cells also play a role in RAWM task performance, we injected scid mice i.v. with a purified population of $CD3^+$ T cells (isolated from genetically identical donors) and tested their performance 2 weeks later. Relative to naive scid mice, the T cell-injected scid mice showed significant improvement both in escape latency (Fig. 3C) and in the number of errors (Fig. 3D).

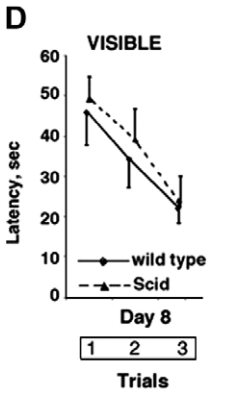
To identify other immune cells that might also play a role in cognition, we replenished another groups of scid mice with T cell-depleted splenocytes or PBS, for control. Relative to PBS-injected scid mice, no changes were observed in mice recipients of T cell-depleted splenocytes either in latency (Fig. 3E) or in the number of errors (Fig. 3F), supporting the contention that T cells constitute the main immune cell population affecting learning behavior.

Although the genetic background of the mice used in our study was identical, the scid mice are characterized by lifelong immune deficiency. It is conceivable that this lack of adaptive immunity could indirectly affect brain development in young mice. With respect to adult mice, we postulated that modulation of the immune system exerts a direct effect on cognitive functions. To examine this hypothesis we depleted WT mice of their immune system by a lethal split-dose gamma-irradiation (Aboud-Pirak et al., 1995), and then immediately injected them with 2×10^6 bone-marrow cells from the genetically matched WT or scid mice. Four weeks after bone marrow transplantation, the immune system was repopulated in mice that received bone marrow of a WT origin, whereas recipients of scid bone marrow had a significant reduction in T cell (CD3) and B cell (CD19) numbers, however there was an increase in myeloid cells (Cd11b; Fig. 4A). Irradiated WT mice replenished with autologous bone marrow showed a learning curve that was typical, both in escape latency and in errors, of naïve WT mice (Fig. 4B and C). In contrast, irradiated WT mice replenished with scid-derived bone marrow (lacking adaptive immunity) exhibited an impaired learning pattern similar to that of scid mice (Fig. 4B and C). These findings demonstrate that depletion of adaptive immunity in adulthood impairs cognitive function. To examine whether cognitive functions of the 'newly formed scid' mice could be restored and to verify that there was no irreversible brain damage as a result of reconstitution with scid bone marrow, we injected one group of 'newly formed scid' mice, 4 weeks after bone marrow transplantation, with autologous T cells obtained from WT donors. Two weeks later we assessed the performance of the RAWM task. Relative to controls, the RAWM performance of T cell-infused 'newly formed scid' mice was significantly improved in both latency (Fig. 4D) and errors (Fig. 4E).

4. Discussion

The results of this study show that adaptive immunity plays a role in mouse learning behavior as examined by spatial learning and memory tasks, such as MWM, Barnes maze, and RAWM. The differences between WT and scid mice were observed primarily during memory acquisition, whereas memory retrieval, as tested in probe trial of





E

Velocity (cm/s)

	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8
w.t.	15.4±1.3	15.1±1.1	12.7±1.2	12.5±2	10.6±2	12.8±3	12.6±1.5	14.3±1.7
scid	15.3±1.8	15.5±1.7	14.3±2.9	10.5±1.5	10.4±1.2	12.8±1	14.2±1	15.1±2
	ACQUISITION				PROBE	REVERSAL		VISIBLE

F

Open Field Activity

	Center crossings, N	Time in center, s	Latency of first center crossing, s	Velocity, cm/s	Distance moved, cm
w.t.	10.4 ± 1.6	10.4 ± 2.8	21.7 ± 10.8	6.7 ± 0.3	2008 ± 89
scid	11.7 ± 2.7	7.7 ± 1.7	47.6 ± 17.9	7.2 ± 0.4	2143 ± 120

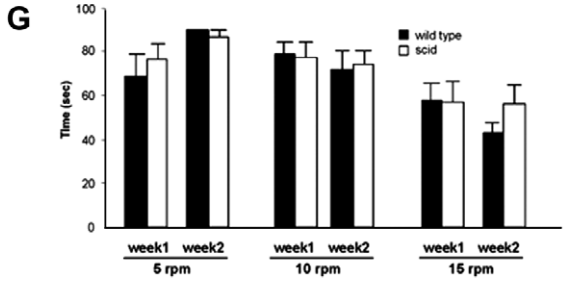


Fig. 1. Learning behavior of scid and WT mice in MWM cognitive task. (A–C) WT and scid mice on identical genetic background were monitored while attempting a spatial learning/memory task in the MWM. During the acquisition (A) and reversal (C) phases of the task, scid mice took significantly longer than wild-type mice to acquire the spatial learning needed. Two-way repeated measures ANOVA was used for statistical analysis, $P < 0.5$ for acquisition and $P < 0.01$ for reversal. Bonferroni post-hoc test was used for individual pair comparison ($^{*}P < 0.05$; $^{**}P < 0.01$; $^{***}P < 0.001$). The presented results are from one of two experiments performed, with 7–9 mice per group in each experiment. Probe trial was performed on the 5th day and the percentage of time spent in the training quadrant was examined. No difference was found between the groups. (D) On the 8th day mice were introduced to MWM with a visible platform. Average escape latencies for three consecutive trials in both groups are presented. (E) All mice were monitored Ethovision (Noldus Information Technology, The Netherlands) video tracking system. Average daily velocities for each group are presented for the duration of the experiment. (F) Moving activity in open field test was registered and analyzed with Ethovision video tracking system. Animals were placed in the center of arena for 5 min and the following parameters were recorded: distance moved, latency of the first entrance in the center of arena, number of center crossings, time spent in center, and average speed. (G) To compare general motor skills of scid and WT mice, animals underwent rotarod task. Data are means ± SEM for duration of walking (seconds) on the revolving rod at different velocities on two consecutive weeks.

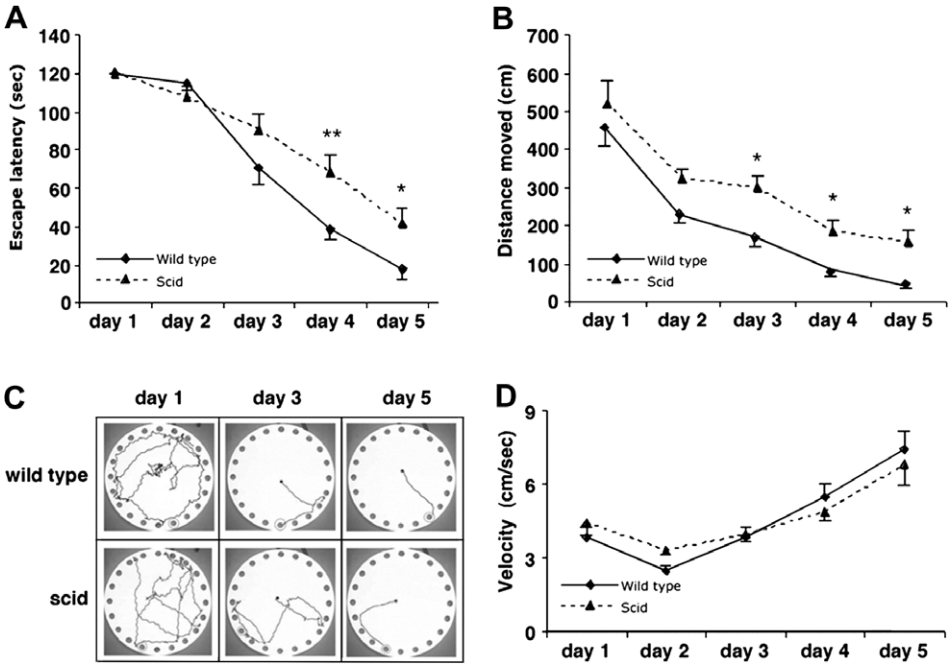


Fig. 2. Scid mice show impaired performance in Barnes maze task as compared to WT counterparts. Scid and WT mice were introduced to Barnes maze. While performing the task animals were monitored and recorded with Ethovision video tracking system. (A) Escape latencies and (B) distance moved to the escape box are shown. Two-way repeated measures ANOVA was used to analyze the data with post-hoc Bonferroni tests used for individual pair comparison. For both latency and distance, $P < 0.05$ ($^{*}P < 0.05$; $^{**}P < 0.01$). (C) Representative tracks are shown. Note the difference in the path length between the groups. On the last day, the scid mouse examined 3 wrong circles on its way to an escape box, unlike the WT mouse that immediately reached the target circle. (D) No difference in average velocities was found between the two groups. One representative experiment is shown out of 2 independent. Each group contained at least 7 animals.

MWM, did not differ between the strains. Furthermore, no difference in motivation or sensori-motor functions was observed in scid mice as compared to WT controls. Acute depletion of adaptive immunity in adult mice led to cognitive decline, which was reversed by injection of autologous T cells from WT donors. These results suggest that T cells participate in memory formation processes and acute changes in adaptive immunity are reflected in learning behavior of mice.

T cells were shown to participate in repair and maintenance of the injured CNS (Hammarberg et al., 2000; Kerschensteiner et al., 2003; Kipnis et al., 2001; Moalem et al., 1999; Wekerle, 2002). T cells also play a role in the healthy CNS, where they support adult neurogenesis (Kipnis et al., 2004; Ziv et al., 2006). The present study revealed significantly impaired performance by scid mice in numerous spatial learning and memory tasks. Velocity of scid mice in different tasks did not differ from that of WT counterparts. Moreover, performance of visible platform test was comparable between WT and scid mice. These results suggest that the difference in task acquisition between the two strains was not as a result of motivation or sensori-motor deficiencies of scid mice.

Our results show that a lack of adaptive immunity is correlated with impaired memory acquisition, as demonstrated by different learning curves in WT and scid mice. However, scid mice did not differ from WT counterparts in a probe trial of MWM suggesting that adaptive immunity does not affect memory retrieval. It is possible that short-term plasticity, which is dependent on post-translational modifications is amenable to control by T cells, whereas long-term plasticity, dependant on gene transcription, is an internally rigorously controlled process, which is not affected by immune cells. Further studies are required to substantiate this hypothesis. Identification of proteins that are involved in short-term plasticity and expression of which is controlled by immune system is a key focus of future studies.

Neurons do not express MHC class II molecules, therefore their direct interaction with T cells is unlikely. Although activated T cells release cytokines that could affect neuronal function (Karni et al., 2006), the state of T cell activation in the CNS following trans-endothelial migration is debatable (Flugel et al., 2001). T cell re-activation in CNS requires their interaction with local antigen presenting cells, primarily microglia and possibly astrocytes (Bailey et al., 2006; Ling et al., 2006; Reinke and Fabry, 2006). Thus, the effect of T cells on learning processes is probably indirect and involves glial cells that upon T cell stimulation acquire a phenotype that affects neuronal function.

T cells can hardly be detected in the CNS of naïve mice and after acquisition of a cognitive task (Chun Hui Yang and Jonathan Kipnis, unpublished observations). These results propose that T cells may affect glial phenotype without entering the parenchyma. One such possibility is that

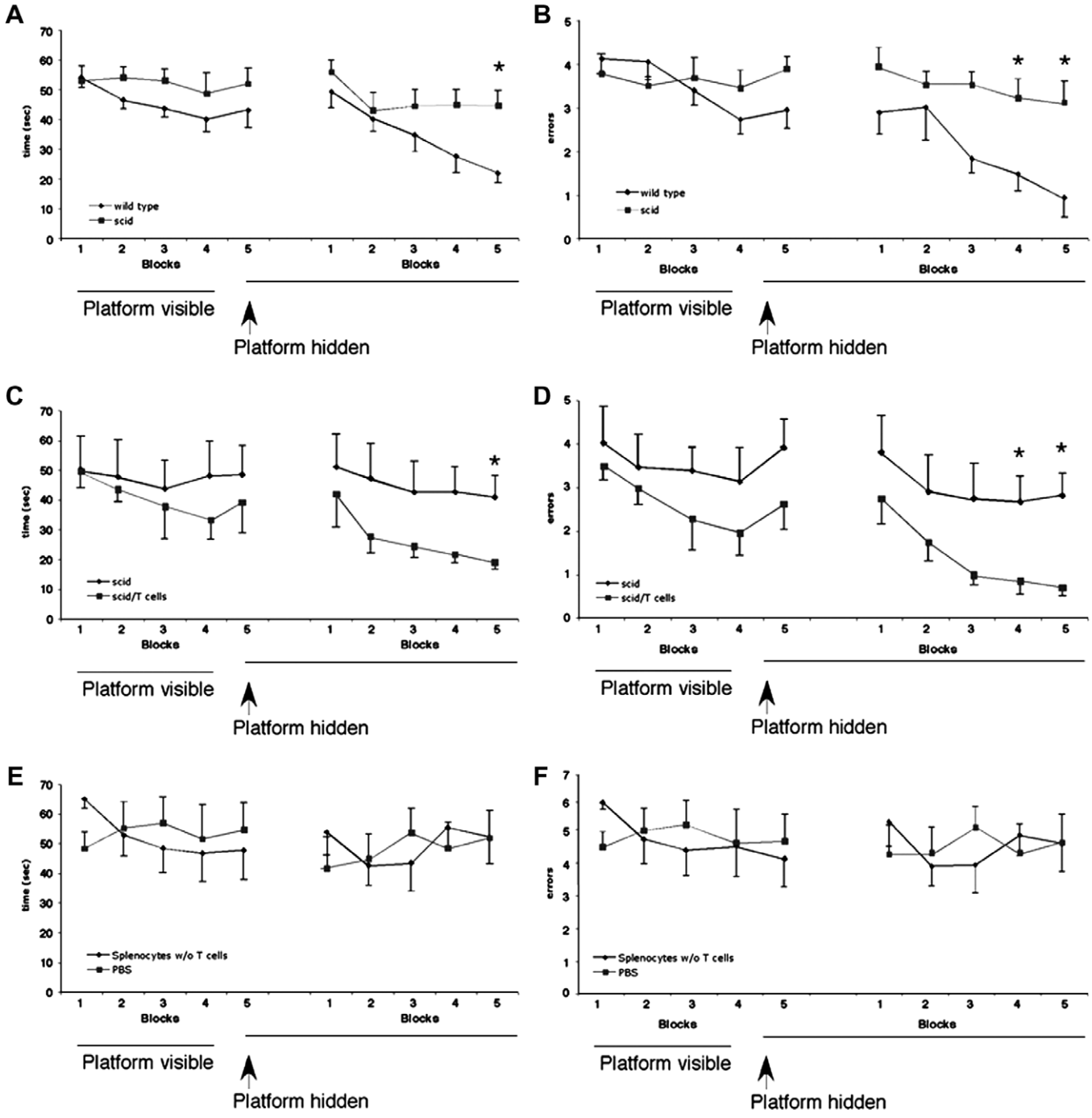


Fig. 3. T cell involvement in performance of RAWM cognitive task. (A and B) Scid mice were compared to genetically-matched WT controls in RAWM task. Time (A) and errors (B) were recorded. Statistical analysis using repeated two-way ANOVA revealed significant impairment in task performance in scid mice as compared to WT controls, in time ($P < 0.05$) and errors ($P < 0.01$). (C and D) Scid mice were injected with either autologous T cells or PBS and three weeks after injection examined in RAWM. Time (C) and errors (D) were recorded. Statistical analysis using repeated two-way ANOVA revealed significant improvement in task performance in scid mice injected with T cells as compared to PBS-injected control mice, in time ($P < 0.05$) and errors ($P < 0.05$). (E and F) Scid mice were injected with splenocytes depleted of T cells and control group was PBS-injected. Three weeks after injection mice were examined in RAWM. Time (E) and errors (F) were recorded. Statistical analysis using repeated two-way ANOVA did not reveal any significant difference between the groups ($P = 0.88$ for time; $P = 0.99$ for errors). Bonferroni post-tests were used to compare individual points. Asterisk (*) corresponds to $P < 0.05$. One representative experiment is shown out of 3 independent. Each group contained at least 8 animals.

astrocytes, composing blood–brain barrier (BBB), may receive signals from T cells across the barrier through interactions of T cells with brain vascular endothelial cells. Another possibility is that T cells interact with mononuclear phagocytic cells in the perivascular spaces. This interaction will lead to T cell activation and production of cytokines, which in turn might affect cognitive function (Chen et al., 2007; Sparkman et al., 2006).

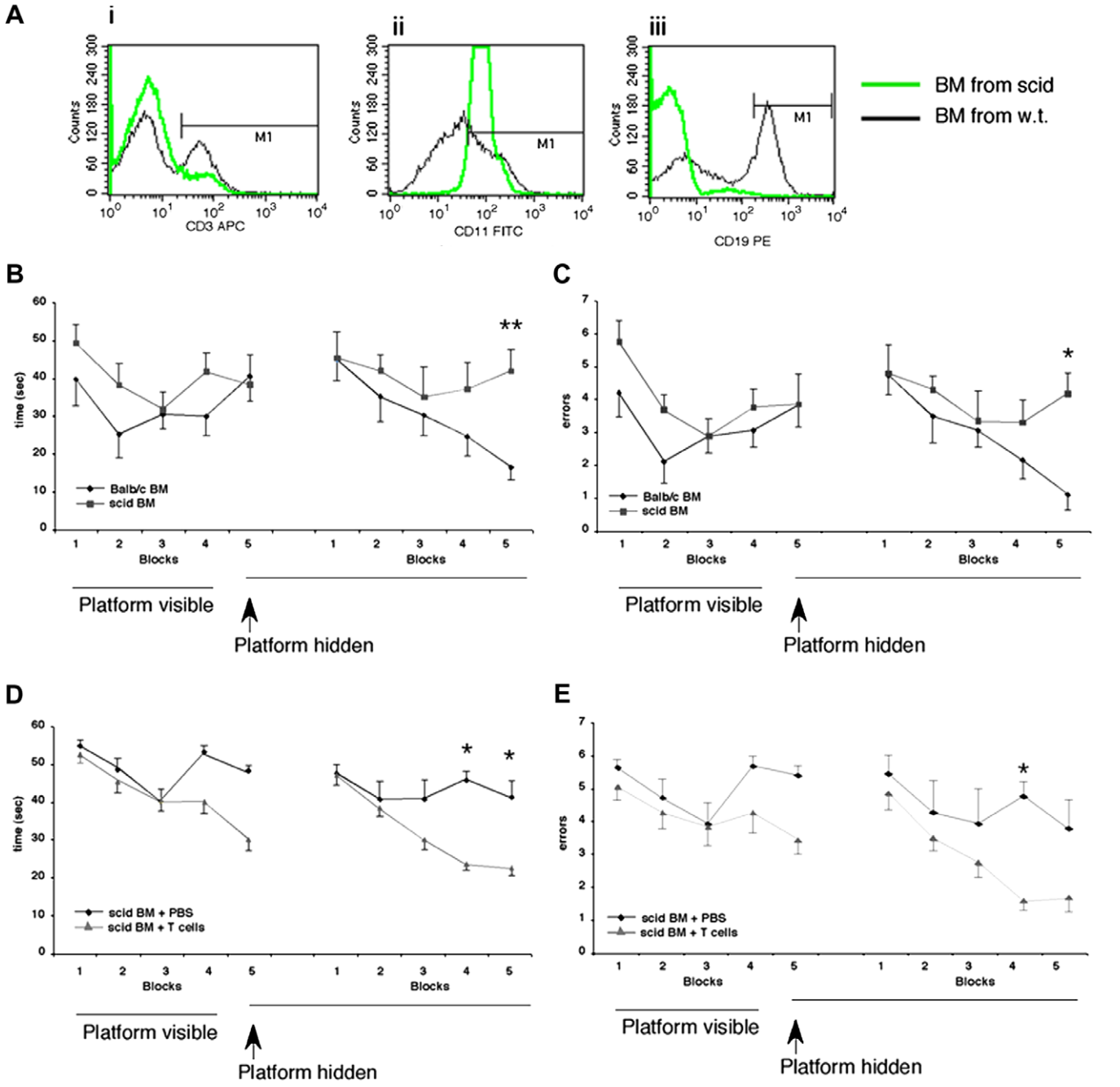


Fig. 4. Depletion of adaptive immunity in adult mice impairs cognitive performance in RAWM and can be restored by passive transfer of T cells. (A) Mice were injected with bone marrow cells from WT or scid origin after a lethal split-dose irradiation. Restoration of T cells (CD3), B cells (CD19) and myeloid cells (CD11b) was examined four weeks following bone marrow reconstitution. (B and C) One month after scid or WT bone marrow reconstitution mice were examined in RAWM. Graphs represent comparison of groups in time latency (B) and errors (C). Statistical analysis using repeated two-way ANOVA revealed significant differences between the groups in time ($P < 0.001$) and errors ($P < 0.01$). (D and E) Another group of WT mice after split-dose irradiation was reconstituted with scid-derived bone marrow and one month later animals were divided into two groups—one injected with PBS and the other with autologous T cells from the WT donors. Two weeks after T cell injection mice were examined in RAWM. Graphs represent comparison of groups in time latency (D) and errors (E). Statistical analysis using repeated two-way ANOVA revealed significant differences between the groups in time ($P < 0.05$) and errors ($P < 0.01$). Bonferroni post-tests were used to compare individual pairs. Asterisks (*, **) correspond to $P < 0.05$ and $P < 0.01$, respectfully. One representative experiment is shown out of 2 independent. Each group contained at least 7 animals.

The devastating effect of cytokines, produced by innate immune cells in response to infection, on brain function has been demonstrated by several groups (Abraham et al., 2006; Burgess et al., 1998; Dantzer et al., 2008; Godbout et al., 2007; Kelley et al., 2003; Shen et al., 2004). It is, therefore, possible that scid mice are undergoing chronic, low-grade inflammation in the periphery. Reconstitution experiments may reduce such an inflammatory burden and, therefore, prevent cognitive impairment induced by over-activated innate immunity.

This possibility is currently being explored in our laboratory.

Scid mice husbandry does not require any special conditions because these mice have a normal functioning innate immunity. However, the lifelong immune deficiency that characterizes scid mice can potentially indirectly affect brain function and neuronal plasticity, therefore it was important to demonstrate the effect of acute immune depletion on brain function in mature mice. The present study demonstrates that ablation of adaptive immune cells from naïve WT mice leads to impaired cognitive performance as examined on RAWM task. Injection of autologous T cells from WT donors into scid mice reversed cognitive deficit, whereas when immune cells other than T cells were injected (T cell-depleted splenocytes) no effect on cognition was evident. Thus, T cells are the major immune players that are involved in the maintenance of lifelong mouse learning behavior.

Understanding the interactions between T cells and resident brain cells that affect neuronal function may alter our perception of how higher brain functions are regulated, and may allow us to intervene in pathological conditions of nervous system by manipulating the immune system. Development of a T cell-mediated vaccine that improves cognitive performance would be a novel and promising approach for age-related dementias (Butovsky et al., 2006; Frenkel et al., 2005) and other cognitive malfunctions associated with immune decline.

Acknowledgments

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